

NONPARAMETRIC ESTIMATION OF THE MODE OF A MULTIVARIATE DENSITY

M. Samanta

University of Manitoba

Key Words: Mode, Multivariate density, Estimation.

1. Summary

The problem of estimating the mode of a probability density function is a matter of both theoretical and practical interest. This problem was first considered by Parzen (1962) in the univariate situation. He has shown that under certain regularity conditions the estimate of the population mode obtained by maximizing an estimate of the true probability density function is consistent and asymptotically normal. The strongest result in this direction is due to Nadaraya (1965) who has proved that under certain regularity conditions the estimate converges to the theoretical mode with probability one. In this paper we consider the problem of estimating the mode of an unknown multivariate probability density function. We establish conditions under which our estimate of the population mode is strongly consistent and asymptotically normal.

2. Introduction

$X_j = (X_{1j}, X_{2j}, \dots, X_{kj})$, $j=1, 2, \dots, n$, are n independent and identically distributed k -dimensional random variables with an unknown common distribution function $F(\underline{x})$ which is assumed to be absolutely continuous having the density function $f(\underline{x})$ where $\underline{x} = (x_1, x_2, \dots, x_k)$. Let $\phi_1, \phi_2, \dots, \phi_k$ be k univariate probability density functions and $\{h_n\}$ be a sequence of positive numbers converging to zero. We consider an estimate $f_n(\underline{x})$ of $f(\underline{x})$ defined by

$$(1) \quad f_n(\underline{x}) = \frac{1}{n} \sum_{j=1}^n h_n^{-k} \prod_{i=1}^k \phi_i \left(\frac{x_i - X_{ij}}{h_n} \right).$$

It is clear that $f_n(\underline{x})$ can be expressed as

$$(2) \quad f_n(\underline{x}) = \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} h_n^{-k} \left\{ \prod_{i=1}^k \phi_i \left(\frac{x_i - u_i}{h_n} \right) \right\} dF_n(\underline{u})$$

where $\underline{u} = (u_1, u_2, \dots, u_k)$ and $F_n(\underline{u})$ is the sample distribution function defined by

$$F_n(\underline{u}) = \frac{1}{n} \sum_{j=1}^n \prod_{i=1}^k \{I(u_i - X_{ij})\}$$

and $l(x-y) = 1$ for $y \leq x$ and vanishes for $y > x$.

We define the following

$$(3) \quad f^{(j_1, j_2, \dots, j_k)}(\underline{x}) = \frac{\partial^{j_1 + j_2 + \dots + j_k} f(\underline{x})}{\partial x_1^{j_1} \partial x_2^{j_2} \dots \partial x_k^{j_k}}$$

$$(4) \quad f_n^{(j_1, j_2, \dots, j_k)}(\underline{x}) = \frac{\partial^{j_1 + j_2 + \dots + j_k} f_n(\underline{x})}{\partial x_1^{j_1} \partial x_2^{j_2} \dots \partial x_k^{j_k}}$$

$$= \frac{1}{n} \sum_{j=1}^n h_n^{-(k+j_1+j_2+\dots+j_k)} \prod_{i=1}^k \phi_i^{(j_i)} \left(\frac{x_i - X_{ij}}{h_n} \right)$$

$$= \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} h_n^{-(k+j_1+j_2+\dots+j_k)} \left\{ \prod_{i=1}^k \phi_i^{(j_i)} \left(\frac{x_i - u_i}{h_n} \right) \right\} dF_n(\underline{u}).$$

Suppose the density function $f(\underline{x})$ is uniformly continuous in $\underline{x}$. Then there exists a vector $\underline{\theta} = (\theta_1, \theta_2, \dots, \theta_k)$ such that $f(\underline{\theta}) = \max_{\underline{x}} f(\underline{x})$. We assume that $\underline{\theta}$ is unique. We next assume that for each i , $\phi_i(t)$ is continuous in t and $\phi_i(t)$ approaches zero along with $|t|$ going to infinity. Therefore, there exists a k -dimensional random variable $\underline{\theta}_n = (\theta_{1n}, \theta_{2n}, \dots, \theta_{kn})$ such that $f_n(\underline{\theta}_n) = \max_{\underline{x}} f_n(\underline{x})$. If for each i , $\phi_i(t)$ is chosen to be twice differentiable, then

$$f_n^{(1,0,0,\dots,0)}(\underline{\theta}_n) = f_n^{(0,1,0,\dots,0)}(\underline{\theta}_n) = \dots = f_n^{(0,0,0,\dots,1)}(\underline{\theta}_n) = 0.$$

It will be shown (Theorem 1) that under certain regularity conditions $\underline{\theta}_n$ converge to $\underline{\theta}$ with probability one. Furthermore, $(nh_n^{k+2})^{1/2}(\underline{\theta}_n - \underline{\theta})$ converges in distribution to a k -dimensional normal random variable with mean vector $\underline{0}$ and covariance matrix $\underline{C}^{-1} \underline{\Lambda} \underline{C}^{-1}$ where

$$\underline{C} = ((c_{rs}))$$

$$\underline{\Lambda} = ((\lambda_{rs}))$$

$$c_{rs} = \left. \frac{\partial^2 f(\underline{x})}{\partial x_r \partial x_s} \right|_{\underline{x} = \underline{\theta}}$$

$$\lambda_{rs} = \begin{cases} f(\underline{\theta}) \int_{-\infty}^{\infty} \{\phi_r^{(1)}(t)\}^2 dt \{ \prod_{\substack{j=1 \\ j \neq r}}^k \int_{-\infty}^{\infty} \phi_j^2(t) dt \}, & \text{if } r = s \\ f(\underline{\theta}) \{ \int_{-\infty}^{\infty} \phi_r(t) \phi_r^{(1)}(t) dt \} \{ \int_{-\infty}^{\infty} \phi_s(t) \phi_s^{(1)}(t) dt \} \cdot \{ \prod_{\substack{j=1 \\ j \neq r,s}}^k \int_{-\infty}^{\infty} \phi_j^2(t) dt \}, & \text{if } r \neq s \end{cases}$$

and $\underline{C}^{-1}$ is the inverse matrix corresponding to $\underline{C}$ (Theorem 2).

We now state the conditions for which the asymptotic properties of the estimate proposed in this paper will be proved.

- (i) $f^{(j_1 j_2, \dots, j_k)}$ are uniformly continuous for $0 \leq j_1 + j_2 + \dots + j_k \leq 2$.
- (ii) The vector $\underline{\theta}$ defined by $f(\underline{\theta}) = \max_{\underline{x}} f(\underline{x})$ is unique.
- (iii) $\phi_j^{(i)}$ are functions of bounded variation for $j = 1, 2, \dots, k$ and $i = 0, 1, 2$.
- (iv) $h_n = n^{-\delta}$, $0 < \delta < \frac{1}{2k+4}$.

3. Main Results

Lemma 1:

There exists a universal constant $C > 0$ such that for each $n > 0$, $\epsilon_n > 0$ and distribution function F ,

$$P_F \{ \sup_{\underline{x} \in R_k} |F_n(\underline{x}) - F(\underline{x})| > \epsilon_n \} \leq C \exp(-2n\epsilon_n^2).$$

Proof:

For $k = 1$, see Kolmogorov (1941) and Dvoretzky, Kiefer and Wolfowitz (1956). For $k > 1$, see Kiefer and Wolfowitz (1958).

Lemma 2:

Under condition (iv)

$$\lim_{n \rightarrow \infty} \sup_{\underline{x} \in R_k} h_n^{-(k+j_1+j_2+\dots+j_k)} |F_n(\underline{x}) - F(\underline{x})| = 0$$

with probability one for $0 \leq j_1 + j_2 + \dots + j_k \leq 2$.

Proof:

For any $\epsilon > 0$, we obtain by Lemma 1

$$\sum_{n=1}^{\infty} P_F \left\{ \sup_{\underline{x} \in R_k} h_n^{-(k+j_1+j_2+\dots+j_k)} |F_n(\underline{x}) - F(\underline{x})| > \epsilon \right\} \\ \leq \sum_{n=1}^{\infty} C \exp(-2n\epsilon^2 h_n^{2(k+j_1+j_2+\dots+j_k)});$$

By condition (iv) the right hand side of the above inequality is finite. An application of the Borel-Cantelli lemma now completes the proof.

Lemma 3:

Under conditions (i) and (iii)

$$\limsup_{n \rightarrow \infty} \sup_{\underline{x} \in R_k} |E\{f_n^{(j_1, j_2, \dots, j_k)}(\underline{x})\} - f^{(j_1, j_2, \dots, j_k)}(\underline{x})| = 0$$

for $0 \leq j_1 + j_2 + \dots + j_k \leq 2$.

Proof:

It suffices to consider the case when

$$0 \leq j_1 + j_2 \leq 2 \text{ and } j_3 = j_4 = \dots = j_k = 0.$$

Using conditions (i) and (iii) and Fubini's theorem, we obtain by integration by parts

$$\sup_{\underline{x} \in R_k} |E\{f_n^{(j_1, j_2, 0, 0, \dots, 0)}(\underline{x})\} - f^{(j_1, j_2, 0, 0, \dots, 0)}(\underline{x})| \\ = \sup_{\underline{x} \in R_k} \left| \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} h_n^{-k} \prod_{i=1}^k \phi_i\left(\frac{x_i - u_i}{h_n}\right) f^{(j_1, j_2, 0, 0, \dots, 0)}(\underline{u}) d\underline{u} \right. \\ \left. - f^{(j_1, j_2, 0, 0, \dots, 0)}(\underline{x}) \right| \\ \leq \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \sup_{\underline{x} \in R_k} |f^{(j_1, j_2, 0, 0, \dots, 0)}(\underline{x} - \underline{t}h_n) - f^{(j_1, j_2, 0, 0, \dots, 0)}(\underline{x})| \\ \cdot \prod_{i=1}^k \phi_i(t_i) d\underline{t}$$

where $h_n^{-1}(\underline{x} - \underline{u}) = \underline{t} = (t_1, t_2, \dots, t_k)$ and $0 \leq j_1 + j_2 \leq 2$. By hypothesis the integrand in the right hand side of the above inequality is bounded above by an integrable function. By Lebesgue dominated convergence theorem the proof follows.

Lemma 4:

Under conditions (iii) and (iv)

$$\lim_{n \rightarrow \infty} \sup_{\underline{x} \in R_k} |f_n^{(j_1, j_2, \dots, j_k)}(\underline{x}) - E f_n^{(j_1, j_2, \dots, j_k)}(\underline{x})| = 0$$

with probability one for $0 \leq j_1 + j_2 + \dots + j_k \leq 2$.

Proof:

Using condition (iii), we obtain by multiple integration by parts (see Hobson (1927))

$$\begin{aligned} & \sup_{\underline{x} \in R_k} |f_n^{(j_1, j_2, \dots, j_k)}(\underline{x}) - E f_n^{(j_1, j_2, \dots, j_k)}(\underline{x})| \\ &= \sup_{\underline{x} \in R_k} \left| \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} h_n^{-(k+j_1+j_2+\dots+j_k)} \left\{ \prod_{i=1}^k \phi_i^{(j_i)} \left(\frac{x_i - u_i}{h_n} \right) \right\} \{dF_n(\underline{u}) - dF(\underline{u})\} \right| \\ &= \sup_{\underline{x} \in R_k} h_n^{-(k+j_1+j_2+\dots+j_k)} \left| \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \{F_n(\underline{u}) - F(\underline{u})\} \left\{ \prod_{i=1}^k d\phi_i^{(j_i)} \left(\frac{x_i - u_i}{h_n} \right) \right\} \right| \\ &\leq \left\{ \sup_{\underline{x} \in R_k} h_n^{-(k+j_1+j_2+\dots+j_k)} |F_n(\underline{x}) - F(\underline{x})| \right\} \cdot \prod_{i=1}^k \text{Var} \{ \phi_i^{(j_i)} \} \end{aligned}$$

for $0 \leq j_1 + j_2 + \dots + j_k \leq 2$ where $\text{Var} \{ \phi_i^{(j_i)} \}$ stands for the variation of $\phi_i^{(j_i)}$. By Lemma 2 the desired conclusion follows.

Lemma 5:

Under conditions (i), (iii) and (iv)

$$\lim_{n \rightarrow \infty} \sup_{\underline{x} \in R_k} |f_n^{(j_1, j_2, \dots, j_k)}(\underline{x}) - f^{(j_1, j_2, \dots, j_k)}(\underline{x})| = 0$$

with probability one for $0 \leq j_1 + j_2 + \dots + j_k \leq 2$.

Proof:

The proof follows from Lemmas 3 and 4 and the inequality

$$\begin{aligned}
& \sup_{\underline{x} \in R_k} |f_n^{(j_1 j_2, \dots, j_k)}(\underline{x}) - f^{(j_1 j_2, \dots, j_k)}(\underline{x})| \\
& \leq \sup_{\underline{x} \in R_k} |f_n^{(j_1 j_2, \dots, j_k)}(\underline{x}) - E f_n^{(j_1 j_2, \dots, j_k)}(\underline{x})| \\
& \quad + \sup_{\underline{x} \in R_k} |E f_n^{(j_1 j_2, \dots, j_k)}(\underline{x}) - f^{(j_1 j_2, \dots, j_k)}(\underline{x})|.
\end{aligned}$$

We now prove the following

Theorem 1:

Under conditions (i), (ii), (iii), and (iv)

$$\lim_{n \rightarrow \infty} \underline{\theta}_n = \underline{\theta}$$

with probability one.

Proof:

The proof follows from Lemma 5 and the fact that for every $\epsilon > 0$ there exists a $\delta > 0$ such that $|f(\underline{x}) - f(\underline{\theta})| \geq \delta$ if $||\underline{x} - \underline{\theta}|| \geq \epsilon$ (see Theorem 2 of Nadaraya (1965) in case of $k = 1$).

Since $\underline{\theta}$ is the unique mode of $f(\underline{x})$, we have

$$f_{i1}(\underline{\theta}) = \left. \frac{\partial f(\underline{x})}{\partial x_i} \right|_{\underline{x} = \underline{\theta}} = 0, \quad i = 1, 2, \dots, k$$

and the matrix $\underline{C} = ((c_{ij}))$ where

$$f_{ij}(\underline{\theta}) = \left. \frac{\partial^2 f(\underline{x})}{\partial x_i \partial x_j} \right|_{\underline{x} = \underline{\theta}} = c_{ij}, \quad j = 1, 2, \dots, k$$

is negative definite.

Similarly under condition (iii) for every sample sequence

$$f_{in}(\underline{\theta}_n) = \left. \frac{\partial}{\partial x_i} f_n(\underline{x}) \right|_{\underline{x} = \underline{\theta}_n} = 0, \quad i = 1, 2, \dots, k$$

and the matrix $\underline{D}_n(\underline{\theta}_n) = ((d_{ijn}(\underline{\theta}_n)))$ where

$$f_{ijn}(\underline{\theta}_n) = \frac{\partial^2 f_n(\underline{x})}{\partial x_i \partial x_j} \bigg|_{\underline{x} = \underline{\theta}_n} = d_{ijn}(\underline{\theta}_n), i, j = 1, 2, \dots, k$$

is negative definite.

Let $\underline{Q}_{1N}$ be any $k \times k$ random matrix whose elements converge in probability to zero and $\underline{Q}_{2N}$ be any row vector with k random components which are bounded in probability. Expanding each $f_{in}(\underline{\theta})$ around $\underline{\theta}$, we obtain

$$(6) \quad f_{in}(\underline{\theta}_n) = f_{in}(\underline{\theta}) + \sum_{j=1}^k (\theta_{jn} - \theta_j) f_{ijn}(\underline{\theta}_n^*) = 0, i=0, 1, \dots, k$$

where $\underline{\theta}_n^* = \lambda \underline{\theta}_n + (1-\lambda)\underline{\theta}$, $0 < \lambda < 1$.

Using Lemma 5 and the fact that $\underline{\theta}_n^* \rightarrow \underline{\theta}$ in probability equations (6) can be written

$$(nh_n^{k+2})^{1/2}(\underline{\theta}_n - \underline{\theta}) = (nh_n^{k+2})^{1/2} \underline{A}_n(\underline{\theta}) C^{-1} + (nh_n^{k+2})^{1/2} \underline{A}_n(\underline{\theta}) \underline{Q}_{1N}$$

where $\underline{A}_n(\underline{\theta}) = (-f_{1n}(\underline{\theta}), -f_{2n}(\underline{\theta}), \dots, -f_{kn}(\underline{\theta}))$.

It is evident that the asymptotic distribution of $(nh_n^{k+2})^{1/2}(\underline{\theta}_n - \underline{\theta})$ depends on the asymptotic distribution of $(nh_n^{k+2})^{1/2} \underline{A}_n(\underline{\theta})$. The following lemma gives the asymptotic distribution of $(nh_n^{k+2})^{1/2} \underline{A}_n(\underline{\theta})$.

Lemma 6:

Under conditions (i), (iii) and (iv), $(nh_n^{k+2})^{1/2} \underline{A}_n(\underline{\theta})$ converges in distribution to a k -dimensional normal random variable with mean vector $\underline{0}$ and covariance matrix $\underline{\Lambda}$ where $\underline{\Lambda} = ((\lambda_{rs}))$ is as defined in (5).

Proof:

We have

$$f_{in}(\underline{\theta}) = \frac{1}{n} \sum_{j=1}^n V_{ijn}$$

where

$$V_{ijn} = h_n^{-k-1} \phi_i \left(\frac{\theta_i - X_{ij}}{h_n} \right) \prod_{\substack{\ell=1 \\ \ell \neq i}}^k \phi_\ell \left(\frac{\theta_\ell - X_{\ell j}}{h_n} \right).$$

Hence,

$$(nh_n^{k+2})^{1/2} \underline{A}_n(\underline{\theta}) = n^{-1/2} \sum_{j=1}^n \underline{Z}_{jn}$$

where

$$\{Z_{jn} = h_n^{\frac{k+1}{2}} (V_{1jn}, V_{2jn}, \dots, V_{kjn}), \quad j=1, 2, \dots, n\}$$

is a sequence of independent and identically distributed k -dimensional random variables.

We define for $r, s=1, 2, \dots, k$

$$\xi_{rn} = E \{ h_n^{\frac{k+1}{2}} V_{rjn} \}$$

$$\sigma_{rsn} = \text{Cov} \{ h_n^{\frac{k+1}{2}} V_{rjn}, h_n^{\frac{k+1}{2}} V_{sjn} \}$$

$$\rho_{rn}^3 = E \{ h_n^{\frac{k+1}{2}} |V_{rjn} - E V_{rjn}| \}^3.$$

Using conditions (i) and (ii), we get

$$E \{ h_n^{(m-1)k+m} |V_{rjn}|^m \} \rightarrow f(\underline{\theta}) \left\{ \int_{-\infty}^{\infty} |\phi_r(1)(t)|^m dt \right\} \left\{ \prod_{i=1}^k \int_{-\infty}^{\infty} \phi_i^m(t) dt \right\}$$

$i \neq r$

and

$$h_n^{k+2} E \{ V_{rjn} V_{sjn} \} \rightarrow f(\underline{\theta}) \cdot \lambda_{rs}.$$

Hence,

$$\xi_{rn} \rightarrow 0$$

$$\sigma_{rsn} \rightarrow f(\underline{\theta}) \cdot \lambda_{rs}$$

and

$$h_n^{\frac{k}{2}} \rho_{rn}^3 \rightarrow f(\underline{\theta}) \left\{ \int_{-\infty}^{\infty} |\phi_r(1)(t)|^3 dt \right\} \left\{ \prod_{i=1}^k \int_{-\infty}^{\infty} \phi_i^3(t) dt \right\}.$$

$i \neq r$

This implies that

$$n^{-\frac{1}{2}} \rho_{rn}^3 = O(nh_n^k)^{-\frac{1}{2}}$$

which by virtue of condition (iv) tends to zero for every r . We now invoke Bernstein's multivariate central limit theorem (see Hoeffding and Robbins (1948)) to complete the

proof of the lemma.

We are now in a position to prove the following

Theorem 2:

Under conditions (i), (ii), (iii) and (iv), $\underline{\theta}_n$ is asymptotically normally distributed with mean vector $\underline{\theta}$ and covariance matrix $\underline{C}^{-1} \Delta \underline{C}^{-1} / n h_n^{k+2}$.

Proof:

Using Lemma 6 relation (7) can be written

$$(n h_n^{k+2})^{1/2} \{ \underline{\theta}_n - \underline{\theta} \} = (n h_n^{k+2})^{1/2} \Delta_n(\underline{\theta}) \underline{C}^{-1} + O_2 N O_1 N.$$

Another application of Lemma 6 completes the proof.

We note that for $k = 1$, Theorem 2 reduces to Theorem 5A of Parzen (1962).

ACKNOWLEDGEMENT

This research has been partially supported by a grant from the National Research Council of Canada.

REFERENCES

- DVORETZKY, A., KIEFER, J. and WOLFOWITZ, J. (1956). Asymptotic minimax character of the sample distribution function and of the classical multinomial estimator. *Ann. Math. Statist.* 27, 642-669.
- HOBSON, E.W. (1927). *The theory of functions of a real variable and the theory of fourier series*. Vol. 1, Cambridge University Press.
- HOEFFDING, W. and ROBBINS, H. (1948). The central limit theorem for dependent random variables. *Duke Math. J.*, 15, 773-780.
- KIEFER, J. and WOLFOWITZ, J. (1958). On the deviations of empiric distribution function of vector chance variables. *Trans. Am. Math. Soc.*, 87, 173-186.
- KOLMOGOROV, A. (1941). Confidence limits for an unknown distribution function. *Ann. Math. Statist.*, 12, 461-463.
- NADARAYA, E.A. (1965). On nonparametric estimates of density functions and regression curves. *Th. Prob. Appl.*, 10, 186-190.
- PARZEN, E. (1962). On estimation of a probability density function and mode. *Ann. Math. Statist.*, 33, 1065-1076.

Manuscript received 4 May 1973